MCS-101	ADVANCED COMPUTER ARCHITECTURE	L T P 3 1 0	4 Credits

Course Objectives:

CO1	To make students know about the Parallelism concepts in Programming
CO2	To give the students an elaborate idea about the different memory systems and buses.
CO3	To introduce the advanced processor architectures to the students.
CO4	To make the students know about the importance of multiprocessor and multicomputers.
CO5	To study about data flow computer architectures

MODULE I

SIMD, MIMD models of parallel processing, classification of parallel computing structure,

MODULE II

High performance memory system, pipelined computer systems, processor architecture for parallel processing, vector Processing,

MODULE III

RISC AND CISC processors, distributed memory/shared architecture.

References

1. *Kai Hwang, "Advanced Computer Architecture," McGraw-Hill.*
2. *Hwang and Briggs, "Computer Architecture and Parallel Processing," McGraw Hill.*

Course Outcome:

CO1	Demonstrate concepts of parallelism in hardware/software.
CO2	Discuss memory organization and mapping techniques
CO3	Describe architectural features of advanced processors.
CO4	Interpret performance of different pipelined processors.
CO5	Explain data flow in arithmetic algorithms